# MZUZU UNIVERSITY

## FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

## ICT 4701: SOFTWARE ENGINEERING

### END OF SEMESTER EXAMINATION

**DATE: DECEMBER, 2015** **TIME ALLOWED: 3 HOURS**

## INSTRUCTIONS

1. Answer **ALL five** questions.
2. This paper contains **three** pages. Please check.
3. Start answering each question on a new page.
4. Write your answers clearly.

**Question 1**

a. Discuss the following Software process models:

i. Waterfall **(2 marks)**
ii. Evolutionary **(2 marks)**
iii. Re-use development **(2 marks)**
iv. Spiral **(2 marks)**

b. Discuss **three** strategic options for legacy system evolution. **(6 marks)**

c. Using a diagram discuss software evolution process. **(3 marks)**

**Question 2**

a. Discuss the following types of software maintenance.

i. Corrective. **(2 marks)**
ii. Adaptive. **(2 marks)**
iii. Perfective. **(2 marks)**
iv. Preventive. **(2 marks)**

b. List **two** reasons why the productivity rate of programmers working as a pair is roughly the same as two programmers working individually. **(2 marks)**

c. Give **two** circumstances where you might recommend against software reuse.

**(4 marks)**

d. Explain how both the waterfall model and the prototyping model can be accomplished in the spiral process model. **(4 marks)**

**Question 3**

a. Discuss **three** disadvantages of using LOC for software costing. **(6 marks)**

b. Discuss any **three** Lehman's laws. **(6 marks)**

c. Explain **three** reasons why program inspections are an effective technique for discovering errors in a program.
**( 6 marks)**

d. Explain the differences between verification and validation, and explain why validation is a particularly difficult process. **(4 marks)**

e. What **three** types of errors are unlikely to be discovered through inspections?
**(6 marks)**

**Question 4**

a. Discuss **four** Software maintenance cost factors. **(8 marks)**

b. Explain any **two** reasons why interface testing is necessary even when individual components have been extensively validated through component testing and program inspections. **(6 marks)**

c. Explain why testing can only detect the presence of errors, not their absence.
**(4 marks)**

d. Discuss risk management process. **(4 marks)**

**Question 5**

**Table 1.**

| Activity | Predecessor | Time Estimates | | | Expected Time |
|---|---|---|---|---|---|
| | | Opt. (O) | Normal (M) | Pess. (P) | |
| A | -- | 2 | 4 | 6 | |
| B | -- | 3 | 5 | 9 | |
| C | A | 4 | 5 | 7 | |
| D | A | 4 | 6 | 10 | |
| E | B,C | 4 | 5 | 7 | |
| F | D | 3 | 4 | 8 | |
| G | E | 3 | 5 | 8 | |

| Key |
|---|
| M = Most Likely Time |
| O = Optimistic Time |
| P = Pessimistic Time |

**Note** that duration is in days.

Refer to **Table 1** to answer the following questions:

a. Compute expected time for each activity. **(7 marks)**

b. Draw network diagram for activities in **Table 1**. **(3 marks)**

c. Find critical path and calculate total time for project completion. **(2 marks)**

d. Calculate slack time for activities E, G and D. **(4 marks)**

**END OF QUESTION PAPER**

*Happy Christmas holidays and prosperous new year*